Optimization Exercises

Practice computation graphs, automatic differentiation, and gradient descent. Show your intermediate steps. Use radians and round numerical answers to six decimal places.

Exercise 1 Forward mode automatic differentiation

$$h(w) = \ln(w) + e^{w}\sin(w), \qquad w > 0$$

a) Draw the computation graph and label every intermediate node.

b) Propagate the function value and derivative at each node, starting with $\dot{w} = 1$.

c) Evaluate all node values and derivatives at $w = 1$.

Exercise 2 Reverse mode automatic differentiation

$$f(x, y) = (x - y)(x \cdot y), \qquad x = 3, \qquad y = 2$$

a) Draw the computation graph, including every intermediate node.

b) Perform the forward pass and compute the value of f.

c) Perform the reverse pass step by step. Sum all contributions to each input to obtain $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Exercise 3 A function with two stages

$$z = \tanh(xy - 1), \qquad x = 1, \qquad y = 2$$

a) Build a computation graph using separate product, shift, and tanh nodes.

b) Perform the forward pass and compute z.

c) Perform the reverse pass to obtain $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

Exercise 4 Gradient descent in one variable

$$g(w) = (w + 2)^2, \qquad w_0 = 2, \qquad \alpha = 0.1$$

a) Write the gradient descent update rule.

b) Compute the first five iterations and the cost at each iteration.

c) Plot or sketch the cost from iteration 0 to 5. Do the iterates approach the true minimum?

d) Bonus: explain what changes for $\alpha = 0.01$ and $\alpha = 1$.

Exercise 5 Gradient descent in two variables

$$g(w_1, w_2) = w_1^2 + 3w_2^2, \qquad (w_1^{(0)}, w_2^{(0)}) = (2,1), \qquad \alpha = 0.1$$

a) Compute the gradient and write the update for each coordinate.

b) Compute the first three simultaneous updates explicitly.

c) Which coordinate shrinks faster in relative terms? Explain why.